

Sandesh S. Kalantre

Knight-Hennessy Fellow, Stanford University | [Website](#) | [Google Scholar](#)

EDUCATION

- Stanford University**, GPA: 4.0/4.0 *2022 - 2026 (exp.)*
PhD in Physics, Knight-Hennessy Fellow, Advisor: Prof. David Goldhaber-Gordon
- University of Maryland (UMD), College Park**, GPA: 4.0/4.0 *2018 - 2022*
Lanczos Graduate Fellow (*transferred to Stanford*)
- Indian Institute of Technology (IIT), Bombay - India**, GPA: 9.87/10.0 *2014- 2018*
B.Tech in Engineering Physics with Honors and Minor in CS, Institue Silver Medal

RESEARCH SKILLS

Experimental

- Magnetron sputtering growth of superconducting films, e-beam and atomic layer deposition, chemical and plasma etching, SEM imaging, 2D material assembly
- Photo and electron-beam lithography for nanostructured devices including superconducting resonator circuits, Josephson junctions, vdW heterostructures, mesoscopic devices, etc.
- Metal machining, 3D printing
- Cryogenic instrumentation — mK dilution fridges, variable temperature inserts and liquid He dipsticks
- RF reflectometry, VNA microwave characterization, custom mixer circuits and high-speed digitizers for pulsed cQED measurements, lock-in amplifiers

Computational and Design

- numpy, scipy, qutip models for quantum dots, 2d materials, Josephson junctions
- Machine learning (Tensorflow) with emphasis towards physical applications to experiments
- Microwave FEM simulations — Sonnet, Maxwell, HFSS
- Solidwords and FreeCAD for experimental design and assembly, PCB design (KiCad)
- COMSOL for electrostatic and magnetostatic simulations

Communication & Leadership

- Academic grant and technical report writing, student mentoring and project management

PATENTS

- Ray-based Classification Framework for Machine Learning-based Tuning Techniques *2023*
J.P. Zwolak, **S.S. Kalantre**, J.M. Taylor, T.W. McJunkin, US Patent App. 18/028,154

RELEVANT RESEARCH EXPERIENCE

- Geballe Lab for Advanced Materials, Stanford University** June 2022 - Present
Advisor: Prof. David Goldhaber-Gordon
- Developed a microwave resonance setup using NbTiN superconducting resonators for probing spin liquid thin films; built a flip-chip assembly for rapid spin resonance characterization of insulating thin films
 - Designed and implemented an architecture drawing from cQED techniques for superfluid stiffness measurements of van der Waals (vdW) superconductors at GHz frequencies
 - Magneto-transport of topological Kondo insulator SmB₆ films and magnetic topological insulator Eu-InAs (324) nanowires to probe surface states and chiral hinge modes respectively

Quantum Materials Lab (QMDLab), University of Maryland

August 2018 - June 2022

Advisor: Prof. James R. Williams

- Probed superconducting interfaces to topological materials and vdW heterostructures through quasi-DC transport and microwave Shapiro measurements
- Developed a theory to explain Shapiro measurements of graphene Josephson junctions building on results in non-linear dynamics of the driven pendulum problem

National Institute for Standards and Technology - Gaithersburg

May 2017 - July 2017

Advisor: Prof. Jacob M. Taylor

- Developed novel machine learning techniques with deep and convolutional neural networks to characterize charge configurations of semiconductor quantum dots
- Engineered a “auto-tuning” algorithm to automatically achieve a desired configuration of quantum dot states; this was used to automatically tune a in-situ device to the single-electron regime and was the first demonstration of such approaches in the field

TOP PUBLICATIONS

Google Scholar: 12 publications and over 500 citations, h-index = 10, i10-index = 10

1. **S.S. Kalantre**, et. al. and J.M. Taylor, Machine learning techniques for state recognition and auto-tuning in quantum dots, *Nature Partner Journal (npj), Quantum Information* 5 (1), 1-10, 2019
2. **S.S. Kalantre**, et. al. and J.R. Williams, Anomalous Phase Dynamics of Driven Graphene Josephson junctions, *Physical Review Research* 2 (2), 023093, 2020
3. C.J. Trimble, M.T. Wei, N.F.Q. Yuan, **S.S. Kalantre** et. al. and J.R. Williams, Josephson Detection of Time Reversal Symmetry Breaking Superconductivity in SnTe Nanowires, *Nature Partner Journal (npj), Quantum Materials* 6 (1), 1-6, 2021
4. M. Pendharkar, S.J. Tran, G. Zaborski Jr, J. Finney, A.L. Sharpe, R.V. Kamat, **S.S. Kalantre**, et. al. and David Goldhaber-Gordon, Torsional force microscopy of van der Waals moirés and atomic lattices, *Proceedings of the National Academy of Sciences* 121 (10), e2314083121, 2024

HONORS AND AWARDS

- Knight-Hennessy Fellowship (\$60,000/yr) 2023-2026
For visionary & collaborative leaders addressing complex real-world challenges at Stanford
- Lanczos Graduate Fellowship 2018-2020
Awarded to top incoming 2-3 students (\$36,000/yr) at Joint Quantum Institute, UMD
- Institute Silver Medal for graduating at the top of Engineering Physics program, IIT Bombay 2018
- S. N. Bose Scholarship (\$2000) awarded to top 50 students from India in STEM 2017
- ISTernship at IST Austria (\$2500) awarded to around 40 students worldwide 2016

TEACHING AND LEADERSHIP EXPERIENCE

Teaching Assistant, Stanford University

2023-2025

- TA for undergraduate E&M, quantum mechanics II; mentored student teams in the design of a nuclear quadrupole resonance setup & cryogenic probes for advanced undergraduate physics lab

Organizational Volunteer and Board Member, USAAAO

2018-2024

- Designed astronomy and astrophysics problems at the high-school level, and assisted in coaching the US national team to the International Olympiad on Astronomy and Astrophysics (IOAA)